Supplementary S4 — Domain-derivation (crosswalk) protocol and coding instructions

Written: 2026-10-25 (UTC), before any item was coded. Coder 1: Claude (Anthropic), model ID `claude-opus-5-5`, working as the `crosswalk` agent (task T5). Coder 2: as first planned, one author independently (see §7); changed by Amendment 1 to the rules (see Addendum C).

Frozen rules: `evidence/crosswalk_rules.md` (sha256 ffe9bd716970122d7e83d5e0086c33a5392612833059f939c80c6adb88d47fea, frozen 2026-10-24, verified unchanged on 2026-10-24/25 before this protocol was written). This protocol does not change the rules. It records (a) the extraction scope, (b) how two undefined terms in the rules are operationalised, (c) the exact coding instructions. Any point below that a reader judges to go beyond an operationalisation should be treated as a candidate dated amendment to the rules file (flagged to the main session).

1. Sources and cached texts

ID	Source	Group	Text used	Item unit
S1	CFQT v3.0 content map (Greinert & Müller 2025, doi 10.2759/8917117); structure as in Greinert et al. 2023	workforce	a local text extraction of the published CFQT v3.0 PDF, pp. 5–13	the 42 subdomains (p. 5); topics of pp. 6–13 kept in `extra`
S2	NSF/NQCO (2020) Key Concepts for Future QIS Learners	school (K-12)	a local text extraction of the published NSF PDF, pp. 2–5	9 Key Concepts + 27 supporting fundamentals (1a–9c). Introduction/ background text excluded. The report states it "is not a consensus document".
S3	Krijtenburg-Lewerissa et al. (2019), IJSE 41(3)	school	a local text copy of the published article; Table 6 column assignment checked against the publisher PDF (layout-preserving text extraction of p. 358)	84 topics of Table 6 (round 2, N = 12); round-3 mean rank (Table 7) in `extra`
S4	Merzel et al. (2024), EPJ QT 11:27	school	the publisher's open-access online article (HTML text, Tables 2 and 5 as HTML, Tables 6 and 8a–8i as images), retrieved 2026-10-25	36 exemplary key concepts of Table 2 (with their category) + 13 illustrations named in the text (§5.3.2, §6.1, §6.3, §6.4)
S5	Bitzenbauer et al. (2025), PRPER 21:020157	school	the CC-BY publisher PDF (institutional repository copy)	105 topics of Tables III–V (round 3, N = 68) with Cns(X) and Include (%)
S6	Stadermann et al. (2019), PRPER 15:010130	school	a local text copy of the published article	17 curriculum items Q1–Q17 (Table VI, countries of 15, documents of 23) + 6 NOS/history aspects N1–N6 (Table I; frequencies from §IV.D.1)

Not extracted at first (limitation, remedied in Addendum B): S4 Tables 7a–7h, 8a–8i, 9a–9b (images); S3 Table 4 (a subset of Table 6, flagged in `extra`); S5 Table I categories are recorded as a category structure (§3), not as items.

Verbatim rule: items are copied exactly, including source typos (e.g., CFQT "communitcation", S5 "0,76", "Technolgies"); parsed values go into `extra`. An automated check (`tools/crosswalk/01_extract.py`) confirms that each verbatim string occurs in the cached source text (whitespace- and hyphenation-normalised).

2. Operationalising "supported by items from" (R-retain) — two readings, both reported

The frozen rules do not define *support*. Both readings below are fixed now, before coding, and R-retain is applied under both. A domain whose retention differs between the readings is reported as **fragile**.

- **Primary (literal) reading.** A source supports a candidate domain if at least one item extracted from that source is coded into it. An item is extracted when the source names it as candidate content for learners at that level (S1: workforce content map; S2–S6: secondary/K-12).
- **Sensitivity (endorsement) reading.** Only items that meet the source's *own* inclusion criterion count:
- S1: all 42 subdomains (S1 cannot create a domain alone in any case).
- S2: all 36 items (a curated list without ratings).
- S3: topics selected by ≥ 8 of 12 experts in round 2 (the authors' cut-off for round 3; 37 topics).
- S4: all 36 exemplary key concepts (nominated by respondents); illustrations only if among the 8 prominent ones (≥ 6 mentions, the authors' cut-off).
- S5: the 10 topics of Table II that met both inclusion criteria (Cns ≥ 0.75 and ≤ 25 % "not needed").
- S6: all 17 Q-items (the authors' list criterion: mentioned in more than one curriculum document) and all six NOS aspects (present in curricula); core-curriculum status (Q3–Q9) is recorded in `extra`.

"School-level support" in R-merge and the "≥ 2 school-level sources" condition of R-retain use the same reading as the count they belong to.

3. Each source's own category structure (verbatim, recorded before coding)

Used only for R-split ("≥ 2 sources treat the parts as separate categories") and R-crosscut ("≥ 2 sources list it as a separate category"). Never used as coding bins.

- **S1 (CFQT, p. 5):** blocks "Quantum background" (domains 1–2), "Core device technologies" (3–4), "QT systems and applications" (5–8). Domains: 1 "Concepts and foundations"; 2 "Physical foundations of quantum technologies"; 3 "Enabling technologies and techniques"; 4 "Quantum hardware"; 5 "Quantum computing and simulation"; 6 "Quantum sensors and imaging systems"; 7 "Quantum communication and networks"; 8 "Valorisation". Proficiency areas (p. 3): "(I) Quantum concepts, incl. mathematical formalism & physics; (II) QT hardware (HW) & software (SW) engineering …; (III) QT applications & strategies … as well as impact, ethics and education".
- **S2 (NSF, p. 1):** "The first Key Concept defines the field as a whole. Concepts 2-6 introduce ideas that are necessary for building an understanding of quantum information science and its applications. Concepts 7-9 provide short explanations of critical areas of study within QIS: quantum computing, quantum communication and quantum sensing."
- **S3 (Krijtenburg-Lewerissa, p. 356–357, Table 5):** round-1 categories "wave–particle duality, wave functions, atoms, subatomic particles, materials, nonlocality and history"; rounds 2–3 grouping (Tables 6–7): "concepts, examples and applications".
- **S4 (Merzel, Table 5):** four classes "Fundamental concepts", "Phenomena and applications", "Mathematical representations", "Philosophical aspects"; 19 KC categories (Table 2); and the distinction between key concepts and illustrations. *CFQT dependence:* respondents were shown the CFQT "as a means to achieve some degree of uniformity" (§4.2), though not obliged to use it.
- **S5 (Bitzenbauer, Table I and Tables III–V):** three fields "Concepts of QP", "Physical Foundations of Quantum Technologies", "Quantum Technologies 2.0"; 23 round-1 categories, including, under Quantum Technologies 2.0, "Quantum computing", "Quantum sensors and metrology", "Quantum communication", "Enabling technologies", "Hardware for quantum computers and sensors". *CFQT dependence:* the categories were assigned "by the authors, using the CFQT", and the experts were shown three CFQT domains. These category boundaries therefore echo the CFQT and are not independent evidence for a separation that the CFQT also makes.
- **S6 (Stadermann, Table III):** five themes "Fundamental QP principles", "Phenomena and applications", "Atomic theory", "Wave function or other mathematical representations", "Philosophical aspects of QP"; separately, Table I "Aspects of nature of science and history of science in quantum physics" (N1–N6); and core curriculum (Q3–Q9) vs extensions.

4. Coding instructions (coder 1, applied exactly as written)

1. Unit: one row of `crosswalk_extraction.csv`. Read `item_verbatim`; for S1 also read the listed topics in `extra` because the subdomain title alone is too terse. Do not read the CFQT domain/field of S5 items or the class of S4 items in `extra` before choosing the code.
2. Assign **one** short content code (2–6 words) that names what the item is *about* (the phenomenon, object, representation, device, practice or issue), using the source's wording. Normalise only spelling and grammatical form so that the same thing gets the same code across sources (e.g., "Tunnelling", "Tunneling", "Quantum tunnelling" → `tunnelling`). Use constant comparison: before creating a new code, check whether an existing code names the same thing.
3. Prohibited bins: do not use the CFQT domain labels (§3, S1) or the R1 manuscript's nine domain labels (Physical foundations; Mathematical foundations; Supporting technologies; QC hardware & sensors; Quantum computing; Quantum sensing; Quantum communication; Practical skills; Societal context) as codes for items whose own wording does not use those words. A source's own term remains a legitimate code when the item itself uses it (e.g., NSF concept 9 "Quantum sensing uses …" → `quantum sensing`).
4. First-revolution devices (lasers, LEDs, transistors, solar cells, chips, STM, MRI, CCD, flash memory, etc.) keep their own device codes; they are not folded into qubit-platform or "hardware" codes.
5. Experiments named as items (double slit, Stern–Gerlach, Mach–Zehnder, photoelectric effect) are coded by the experiment, not by the principle they are used to illustrate.
6. An item that names several things gets the code of its head concept (the first-named or defining one); the note records what else it contains. Multi-topic CFQT subdomains are coded at subdomain level.
7. Write a ≤ 20-word note when the code required a judgement (ambiguous wording, multi-topic item, a code chosen over an alternative).
8. Disclosure: coder 1 also performed the extraction and had therefore seen every source's tables and category structure before coding; instruction 1 limits, but cannot remove, that exposure. This is why an independent second coder is required (§7).
9. Coder 1 codes all items. Codes are written to `evidence/crosswalk_coding.csv` with `coder = claude-opus-5-5`.

5. Clustering instructions

Group content codes (not items) into candidate domains by **shared subject matter**: codes go together when they are about the same kind of thing (e.g., the same family of phenomena, the same family of devices, the same representational apparatus, the same kind of societal issue). Source category placement is not used. Each code goes to exactly one candidate cluster; codes that fit no cluster with others stay as singletons and become residuals unless retained. The cluster label is written after the members are fixed. Output: `evidence/crosswalk_clusters.csv` with item counts per source under both readings.

6. Rule application

R-retain → R-merge → R-split → R-crosscut, in that order, each decision logged with reasons and source counts in `evidence/crosswalk_decisions.md`. Prerequisite judgements for R-split use placement rule P1 (`evidence/placement_rules.md`). Only after all four rules are applied is the result compared with the CFQT content map, the R1 nine domains and Semerikov et al. (2026).

7. Second coder (independent) and agreement

- Sample: ≥ 40 % of items, stratified by source (ceil(0.4 × n) per source), drawn with Python `random.seed(20261025)`; `random.sample` on each source's item list in item-ID order; sources processed S1…S6.
- Packet `evidence/crosswalk_second_coding_packet.csv` shows only `item_id, source, item_verbatim, locator` and blank `second_code, coder`; `extra` is withheld (it contains CFQT fields/classes that would cue the coder).
- Stage 1: the author writes open codes blind (without opening `crosswalk_coding.csv`, `crosswalk_clusters.csv` or `crosswalk_decisions.md`). Stage 2: after returning stage 1, the same author assigns each of their own codes to one cluster from the cluster list, or "none". Claude does not map the author's codes.

- Agreement: percentage of sampled items assigned to the same cluster by both coders, and Cohen's κ over clusters (incl. "none"). Disagreements are resolved by discussion and logged. Until then the domain set is **provisional**.

Addendum A — 2026-10-25 (added after clustering and rule application, before any second coding)

These points were not in §§5–7 as written before coding. They are recorded here so that the supplementary trace matches what was done.

1. **Residual fit test.** Items in a candidate cluster that fails R-retain are counted as residual unless they would have been clustered with a retained domain's codes under the §5 criterion. Applied this way, no item of a failed cluster was moved into a retained domain.
2. **Purposive add-on to the second-coder packet.** In addition to the stratified random 40 % (138 of 339 items; seed 20261025; method unchanged from §7), the packet contains the "pivotal" items, i.e. those whose recoding would flip an R-retain or R-crosscut result (listed in `evidence/crosswalk_decisions.md` §5). Ten of the 15 were already in the random draw, so five were added. The combined list was shuffled with the same random generator and the added items are not marked. Agreement is reported separately for the random sample and for the 15 pivotal items. The sampling record is in `evidence/crosswalk_second_coding_sample.json`.
3. **Stage 2 uses all 13 candidate clusters plus "none"**, described by subject matter only (`evidence/crosswalk_second_coding_stage2.md`), not the retained domains. Agreement therefore measures coding and clustering, not the rule outcomes.
4. **Packet locators for S4** give "Table 2, row n" instead of the source's category name, so that the source category does not cue the coder. Extraction content is unchanged.

Addendum B — 2026-10-25 (UTC): Merzel et al. Tables 8a–8i added after the first rule application

The first application (339 items) omitted Merzel et al.'s image tables. They were retrieved from the publisher's online article (the PDF was not obtainable), read and transcribed, and 61 distinct illustrations not already extracted were added as S4-050 to S4-110 (400 items in all). These items were coded under §4 without change to the instructions. Their verbatim check is against our own transcription of the images only. Endorsement for them follows §2 (Merzel et al.'s own cut-off of at least 6 mentions and at least 4 KC categories; none qualifies). One item whose content cannot be determined from its wording ("Bernstein 1967") is placed in an unclassifiable cluster that cannot be retained. When R-merge clause 2 applies, the merge partner is the retained domain whose A1/A2 objective the school-level items serve (the clause-1 test); this criterion was written down before the partner was named. The second-coder sample was redrawn from the 400-item set with the same seed and method (162 random items, 40.5 %, plus 8 pivotal items not drawn; 170 in all). Results are in `evidence/crosswalk_decisions.md`, Addendum B.

Addendum C — 2026-10-25 (UTC): inter-model second coding (Amendment 1)

This is an **inter-model check, not a human verification** of the coding. Amendment 1 to the frozen rules (dated 2026-10-25, authors' decision) replaced the author second coder with a second large language model from a different developer.

Model and access. NVIDIA Nemotron 3 Ultra, Ollama model tag `nemotron-3-ultra:cloud`, called through a local Ollama daemon (version 0.30.6) that forwards to Ollama's cloud service (`/api/chat`). Model record from `/api/show` (retrieved 2026-10-25T15:24:29Z): parameter size 550000000000, context length 262144, capabilities completion, thinking, tools. The model's default reasoning ("thinking") was left on; the API returns it in a separate field, which is kept in each response file.

Settings. temperature 0; seed 20261025; stream false; batches of 20 items (last batch smaller) in packet order; output constrained by a JSON schema and validated (every input item exactly once; stage-2 values restricted to K01–K13 or "none"); at most two retries per batch. Run from 2026-10-25T15:24:29Z to 2026-10-25T15:47:23Z. Retries: none.

Input. The 170 packet items (162 random, 40.5 %, plus 8 pivotal items not drawn; unmarked). For each item the model received only its ID, source ID, a one-line source type, the verbatim item text, the locator and, for S1, the topics listed under the subdomain. In stage 1 it also received the list of its own labels from earlier batches. Nothing derived from the first coder (codes, cluster memberships, decisions) was sent. No email address or personal data was sent (the script refuses any request containing "@").

Stage 1 system prompt (verbatim):

```
You are coding items for a research synthesis on quantum physics and quantum technology education. Each item is a topic, concept, competence or content item quoted verbatim from one of six published sources: a European competence framework for the quantum-technology workforce (S1); a US list of key concepts for future learners of quantum information science (S2); an expert Delphi study on key topics for secondary-school quantum mechanics (S3); a multi-stakeholder study of key concepts and illustrations for secondary-school quantum physics (S4); an expert Delphi study on secondary-school quantum physics (S5); and an analysis of secondary-school quantum-physics curricula in 15 countries (S6).

Task (open coding). For each item, write a short label (2-6 words) that names what the item is ABOUT: the phenomenon, object, representation, device, practice or issue. Rules:
1. Use the source's own words where you can. Examples of the kind of label meant (not taken from the items): "Compton scattering", "photon counting statistics", "history of the photon concept".
2. Give the same label to the same thing across sources (for example "Tunnelling", "Tunneling" and "Quantum tunnelling" all get one label). Before inventing a new label, check whether one you have already used fits; your labels from earlier batches are listed in the request.
3. Do not sort items into ready-made categories. In particular, do not use any of the following as a label unless the item itself uses those words: Concepts and foundations; Physical foundations of quantum technologies; Enabling technologies and techniques; Quantum hardware; Quantum computing and simulation; Quantum sensors and imaging systems; Quantum communication and networks; Valorisation; Physical foundations; Mathematical foundations; Supporting technologies; QC hardware & sensors; Quantum computing; Quantum sensing; Quantum communication; Practical skills; Societal context. (An item that says, for example, "Quantum sensing uses ..." may be labelled "quantum sensing".)
4. Name devices as devices (for example "lasers", "solar cells", "MRI"). Name an experiment by the experiment (for example "Stern-Gerlach experiment"), not by the principle it shows.
5. If an item names several things, label its main (first-named or defining) idea. For items from S1, the topics the framework lists under that subdomain are given as context; use them to understand the item.
6. If the content of an item cannot be determined from its wording, use the label "unclear".

Return only JSON of the form {"items": [{"item_id": "...", "code": "..."}]}, with exactly one entry for every input item and no other text.
```

Stage 2 system prompt (verbatim; {CLUSTERS} was replaced by the 13 definitions below, one per line as "ID label: definition"):

```
You are coding items for a research synthesis on quantum physics and quantum technology education. In an earlier stage you gave each item a short label describing what it is about. Now assign each item to exactly one of the candidate clusters listed below, or to "none" if no cluster fits. Decide from the item text and your own label. Choose the cluster whose subject matter the item shares; do not try to guess what any other coder did.

Candidate clusters:
{CLUSTERS}

Return only JSON of the form {"items": [{"item_id": "...", "cluster": "K01"}]}, where cluster is one of K01, K02, K03, K04, K05, K06, K07, K08, K09, K10, K11, K12, K13 or none, with exactly one entry for every input item and no other text.
```

Neutral cluster definitions sent in stage 2 (written from the cluster labels and the rules file, not from the first coder's member codes):

- K01 Quantum principles and phenomena: Fundamental principles of quantum physics and the phenomena and experiments in which they show themselves.
- K02 Mathematical representations: Mathematical descriptions and formalisms used to represent quantum states, their change over time and the outcomes of measurements.
- K03 Atoms, light and matter: Quantum physics of atoms, of light and its interaction with matter, and of the structure and properties of matter and materials.
- K04 Particle and nuclear physics: Subatomic particles, atomic nuclei and the processes they undergo.
- K05 Quantum information and qubits: Information represented and processed in quantum systems, with the qubit as its basic unit.
- K06 Quantum computing and simulation: Computation and simulation performed with quantum systems, including quantum computers, their operations, algorithms and applications.
- K07 Quantum communication and cryptography: Transmission and protection of information using quantum systems, and cryptography in a quantum context.
- K08 Quantum sensing and metrology: Measurement, sensing and timekeeping that exploit quantum systems, and the standards of measurement.
- K09 Qubit hardware platforms: The physical systems and engineering used to realise, control and scale qubits.

- K10 Quantum devices and enabling technologies: Devices whose operation relies on quantum physics, and the technologies and techniques used to build and run quantum technologies.
- K11 Valorisation and societal impact: Economic, business, societal and ethical aspects of quantum technologies and their communication to the public.
- K12 Nature, history and philosophy of quantum physics: How quantum physics developed, how it is interpreted, and what it shows about the nature of science.
- K13 Practices: Activities that learners carry out, such as programming, simulation, experimentation and building.

Records. `evidence/second_coder_llm/`: `batch_001–batch_018` request and response JSON (stage 1 = batches 1–9; stage 2 = the rest), `manifest.json` (item IDs, sha256 of every request and response, UTC times, attempts), `model_info.json`, `run_log.txt`. Output: `evidence/crosswalk_second_coding_llm.csv`.

Batch	Stage	Items	Request sha256 (first 16)	Response sha256 (first 16)
001	1	20	8e162f16960a639c	438eb2a8d4f380a6
002	1	20	138e1bdbc3eece2f	7eeba6d822a8fbf1
003	1	20	3ac5f5c8d7bb372b	ca75d4b2975ff29d
004	1	20	9acb778ecb788ea5	38e1fa3ca46419f8
005	1	20	3b1183af04bb70c2	cf9ba769e9b2baf5
006	1	20	da18eedfd281f744	432b9ebb906e4858
007	1	20	9967588edee31a76	7a2ea5208ca330da
008	1	20	00f72aa432203fa8	aea602a577871ebb
009	1	10	c059a6b1471482da	41f1b530a8d846ed
010	2	20	38f978875fa6b908	33cf6377b664fdc9
011	2	20	c10778866998514d	8ef0eef27be7432a
012	2	20	3a0d973f7c29f94d	97066004ee91c936
013	2	20	b925b0b5afcac51e	6fb572c823cce7a9
014	2	20	ef817315551745b6	8d45a886e6fda42e
015	2	20	a7cae4e0b465d77a	d7cda8fde838ce9f
016	2	20	78e3e6840b853715	a019f39851e3444e
017	2	20	70e78f5401231d6a	12141d09c0e4edbb
018	2	10	12fb396e45311da4	eaa0e3383edd29f6

Agreement on cluster assignment (first coder's cluster for the item vs the second model's stage-2 cluster; the first coder's unclassifiable cluster is compared as "none"):

Set	n	Agreement	Cohen's kappa
Random sample	162	138/162 = 85.2 %	0.826
Pivotal items	14	11/14 = 78.6 %	0.731
All packet items	170	144/170 = 84.7 %	0.822

Disagreements: 26, listed in `evidence/crosswalk_disagreements.csv` for adjudication by the authors (empty `author_decision` column). Full figures, per-cluster counts and the confusion summary: `evidence/crosswalk_agreement_llm.json`.

Retention sensitivity (R-retain recomputed with the second model's cluster substituted for every sampled item):

- literal reading: no retention result changes.
- endorsed reading: no retention result changes.

Disagreements on pivotal items: S3-046 "Schrödinger’s cat" (first coder K12, second model K01); S4-097 "Coherent control" (first coder K09, second model K10); S6-018 "Methodology (e.g., experiments and hypothesis)" (first coder K12, second model K13).

Consequences beyond R-retain. (1) Qubit hardware platforms (K09) stay at three sources on the literal reading with the second model's clusters, but then rest on a single single-mention illustration ("NV-center"), because the second model placed "Coherent control" with devices and enabling technologies. (2) The second model placed Stadermann et al.'s nature-of-science aspect "Methodology (e.g., experiments and hypothesis)" with practices. If the authors adopt that coding, a second source arguably lists practices as a separate category, and R-crosscut would make practices a domain rather than a cross-cutting layer. This is the one disagreement with a possible rule consequence; it is flagged for adjudication. (3) The second model gave both Delphi items "Schrödinger's cat" the same stage-1 label but assigned them to different clusters (principles; nature, history and philosophy), an inconsistency within the second coder.

Conditions on the agreement figures. The stage-2 cluster definitions were written by the first coder (from the cluster labels and the rules file only), and the clusters themselves are the first coder's candidate clusters; the agreement is therefore agreement on assignment to the first coder's cluster set, not an independent clustering. The disagreements concentrate on the boundaries between principles, mathematical representations and atoms/light/matter (mathematical representations → principles 6; atoms, light and matter → principles 5), which bears on the clustering granularity and changes no result.

Combined sensitivity (Merzel et al. and Bitzenbauer et al. counted as one source, because they report the same Delphi study): literal/coder1: K13 retain → fail; literal/coder2_substituted: no change; endorsed/coder1: no change; endorsed/coder2_substituted: K12 retain → fail. Under the primary (literal) reading nothing changes except the practices cluster, which is a cross-cutting layer by R-crosscut in any case; under the endorsement reading with the second model's clusters, the nature, history and philosophy domain would fail.

Until the authors have adjudicated the disagreements, the domain set remains provisional.

Addendum D — 2026-10-26: adjudication of the coder disagreements by a language model

Who decided, and why a model. The authors asked that the disagreements be adjudicated by Claude Opus 5.5 (Anthropic, model `claude-opus-5-5`) and that they check every decision afterwards. The adjudicator ran as a fresh Claude Code subagent at the highest reasoning-effort setting, with no access to the context of the session that coordinated the revision. **It is the same model as coder 1**, so a preference for coder 1's assignments is to be expected. That is why the share of decisions siding with each coder is reported below. The reporting plan was recorded before any decision was seen.

Blinding. - Options A and B appear in the same random order as in the authors' packet (seed 20261026). - The coders' own code labels were removed from the options, because their wording could reveal which coder gave them. - The adjudicator worked in a separate folder that held only the packet, the cluster definitions and the six source texts. The key linking options to coders was held outside the project's evidence folder during the run. - It was told not to infer which coder gave which option, or what a decision would change.

Prompt (verbatim; local path replaced):

```
You are the adjudicator for a coding exercise in an academic study. Your working folder is:
<working folder>/

Read INSTRUCTIONS_FOR_MODEL.md in that folder first and follow it exactly. In short:
- decide all 26 rows of PACKET_FOR_MODEL.csv, using only cluster_definitions.md and the source texts in sources/;
- write DECISIONS_MODEL.csv atomically after every row and append to progress.log;
```

```
- finish with SUMMARY.md.

Rules:
- Work only inside this folder. Do not open, list or search any file or directory outside it, and do not use the web.
- Do not try to infer which coder gave which option, or what a decision would change.
- Quote paths in shell commands, because they contain spaces, '&' and parentheses. Use python3 -I for any script.
- Take the time each row needs. Read each item in its source context before deciding.

When done, reply with:
- the number of rows decided;
- the counts of each decision type;
- the path of DECISIONS_MODEL.csv.

Do not include anything else.
```

The run stopped at the account's usage limit after the adjudicator had read every item but before it saved any decision. It was resumed in the same context with a message asking it to write its decisions row by row as instructed, under unchanged rules. The full run record, with that message verbatim, is published as S4_adjudication_run_records.md.

The adjudicator also had an advisor tool, which Claude Code provides. The tool forwards the subagent's own transcript to a stronger Anthropic reviewer model and returns advice. The transcript shows three advisor calls. The advice is redacted in the transcript and is not recorded. Because the advisor sees only that transcript, it had no information from outside the adjudication folder; it may, however, have influenced some decisions.

Instructions file given to the adjudicator (verbatim):

```
# Adjudicating coding disagreements (instructions for the model adjudicator)

**Task.** Two coders assigned topic items, taken verbatim from six published sources, to a fixed set of clusters. For the 26 items in `PACKET_FOR_MODEL.csv` they chose different clusters. Decide each item.

**Files you may use. Work only inside this folder (`adjudication_run/`).**
- `PACKET_FOR_MODEL.csv`: one row per item, in random order. `option_A` and `option_B` are the two coders' clusters. Which coder gave which option varies at random from row to row.
- `cluster_definitions.md`: the definitions. These are the only basis for a decision.
- `sources/`: the six source texts (S1 to S6, named by file) and `all_extracted_items.csv`. That file lists every item extracted from the six sources, with its locator, so you can see the table, column or list in which an item appears and its neighbouring items.

**Do not open, list or search anything outside this folder.** That covers the parent directory and its siblings (evidence/, tools/, agents/, source/, backup/, drafts/, supplementary/, and the state and plan files), and also the web. Do not try to work out which coder gave which option, whether an item matters for any later result, or what any decision would change. None of that is a basis for a decision.

**How to decide each row.**
1. Read the item where it appears in its source. Use the `locator` column, the source text, and the neighbouring items in `all_extracted_items.csv`.
2. Ask which cluster definition the item's subject matter fits best, as it would be taught or discussed at secondary level.
3. In `decision`, write one of:
   - `A`
   - `B`
   - `other: K##`, if a different cluster fits better than both options;
   - `none`, if no cluster fits.
4. In `reason_one_line`, give one line of at most 30 words that refers to the definition. For example: "about how measurement changes the state, so K01 not K02".
5. In `confidence`, write `high`, `medium` or `low`.
```

```
**One item may look ambiguous:** a "Methodology (e.g., experiments and hypothesis)" entry from a curriculum comparison's nature-of-science table. Decide it like any other row. The question is whether the entry is about a student practice (doing experiments) or about knowledge of how science works (the nature of scientific methods). Use the source's own table context.

**Saving (required, because the session may stop at any time).**
- After deciding each row, rewrite the whole of `DECISIONS_MODEL.csv`. It has the packet's columns plus `confidence`. Write it atomically: write `DECISIONS_MODEL.csv.tmp`, then rename it, for example with `python3 -I`.
- Then append one line to `progress.log`: UTC time, row number, decision.
- If `DECISIONS_MODEL.csv` already exists when you start, keep its decided rows and continue with the rest.
- Never modify `PACKET_FOR_MODEL.csv`.

**When all 26 rows are decided**, write `SUMMARY.md`. It should give:
- the number of each decision type (A, B, other, none);
- the rows with `low` confidence;
- any row where neither option fits well, with a one-line explanation.
```

Decisions (unblinded).

Item	Source	Item text	Coder 1	Coder 2	Final	Sided with	Confidence	Reason (adjudicator)	Authors' check
S1-011	S1	Computers and software	K13	K10	K10	coder 2	high	CFQT places it under Enabling technologies and techniques: IT infrastructure and control software used to run QT, matching K10, not a learner activity (K13).	agree
S2-016	S2	At a particular moment in time, a set of n classical bits can exist in only one of 2n poss	K05	K01	K05	coder 1	high	Compares n classical bits with n qubits for storing and processing information; matches K05's information represented and processed with qubits, not a general K01 principle.	agree
S3-026	S3	Stationary states	K02	K01	K02	coder 1	low	A stationary state is defined by how a quantum state changes over time (energy eigenstate, time-independent solution), matching K02's 'change over	agree

Item	Source	Item text	Coder 1	Coder 2	Final	Sided with	Confidence	Reason (adjudicator)	Authors' check
								time'; K01 reading also plausible.	
S3-029	S3	Development of atomic models	K03	K12	K12	coder 2	medium	'Development' of atomic models is the historical sequence of models, showing how quantum physics developed and models' tentativeness: K12; atomic content itself appears as separate items.	agree
S3-030	S3	Free vs. localised particle	K02	K01	K01	coder 2	low	Conceptual contrast between delocalised free and confined particles (quantisation, uncertainty); a fundamental principle (K01) rather than a mathematical formalism (K02).	agree
S3-041	S3	Magnetism	K03	K01	K03	coder 1	medium	An everyday example explained by electron spin; magnetism is a property of matter and materials (K03) rather than a fundamental principle or experiment (K01).	agree
S3-046	S3	Schrödinger’s cat	K12	K01	K12	coder 1	low	Thought experiment on macroscopic superposition and the measurement problem; chiefly a vehicle for discussing interpretation, so K12 rather than an experiment showing principles (K01).	agree

Item	Source	Item text	Coder 1	Coder 2	Final	Sided with	Confidence	Reason (adjudicator)	Authors' check
S3-050	S3	Heat radiation	K03	K01	K03	coder 1	low	Heat (thermal) radiation is light emitted by matter because of its temperature; quantum physics of light and its interaction with matter (K03) fits better than general principles (K01).	agree
S3-081	S3	Single molecule microscopy	K10	K08	K08	coder 2	low	An application that images individual molecules: a measurement and imaging technique exploiting quantum systems (K08), more specific than a generic quantum device (K10).	agree
S4-023	S4	Pauli Exclusion principle	K03	K01	K03	coder 1	medium	Source clusters it under Atomic models (with orbitals, band structure); taught to explain electron shells and periodic table, so K03 over the general-principle reading (K01).	agree
S4-044	S4	Double-well	K02	K01	K01	coder 2	low	Offered as an illustration of Superposition, Quantum measurement, Heisenberg principle and Statistical nature; a system where these K01 principles show themselves, not a formalism (K02).	agree
S4-050	S4	Light interference	K01	K03	K01	coder 1	high	Offered as an illustration of Superposition; interference is a	agree

Item	Source	Item text	Coder 1	Coder 2	Final	Sided with	Confidence	Reason (adjudicator)	Authors' check
								phenomenon where the superposition principle shows itself (K01), not quantum physics of light-matter interaction (K03).	
S4-062	S4	Space and velocity measurement	K01	K08	K01	coder 1	high	Illustration for the Quantum measurement key concept: measuring a particle's position and velocity is about the measurement principle (K01), not sensing technology (K08).	agree
S4-066	S4	Malus’ law with a single photon	K01	K03	K01	coder 1	high	Offered for Quantum measurement and Statistical nature; single-photon transmission probability shows probabilistic measurement, an experiment where K01 principles show themselves, not K03 light-matter physics.	agree
S4-069	S4	Analogy of standing waves and a particle in a box	K02	K01	K01	coder 2	low	Offered as an illustration of Quantization (Table 8c); the standing-wave analogy conveys the quantisation principle conceptually (K01) rather than a mathematical formalism for states (K02).	agree
S4-090	S4	Bloch sphere	K02	K05	K02	coder 1	high	Listed under Mathematical foundations; the	agree

Item	Source	Item text	Coder 1	Coder 2	Final	Sided with	Confidence	Reason (adjudicator)	Authors' check
								Bloch sphere is a representation of a two-level quantum state, so K02 representations, not K05 information processing.	
S4-097	S4	Coherent control	K09	K10	K09	coder 1	medium	Offered for the Qubit key concept; coherent control means manipulating qubit states with pulses, matching K09 'control qubits' more specifically than K10's general enabling technologies.	agree
S5-032	S5	Photoelectric effect	K03	K01	K01	coder 2	medium	Listed among double slit, Mach-Zehnder and matter-wave interference experiments; demonstrates light quanta, an experiment where K01 principles (wave-particle duality) show themselves, over K03.	agree
S5-041	S5	Electrons	K03	K04	K03	coder 1	low	Listed beside spin, Franck-Hertz experiment and atomic models in the quantisation sequence; electrons as constituents of atoms fit K03 better than particle physics (K04).	agree
S5-048	S5	Quantum harmonic oscillator	K02	K01	K02	coder 1	low	A model system whose quantised states follow from the Schrödinger equation; a	agree

Item	Source	Item text	Coder 1	Coder 2	Final	Sided with	Confidence	Reason (adjudicator)	Authors' check
								mathematical description of quantum states (K02) rather than a principle or experiment (K01).	
S5-067	S5	Spectroscopy	K03	K08	K03	coder 1	high	Source lists spectroscopy under atoms and atomic physics (Physical Foundations); it concerns light-matter interaction in atoms, K03, not quantum-enhanced measurement (K08).	agree
S5-085	S5	Topological effects	K03	K01	K03	coder 1	high	Physical Foundations, materials/ nanostructures group (CFQT solid-state topic); topological effects are properties of materials, so K03, not a fundamental principle (K01).	agree
S5-086	S5	Surface tension	K03	none	none	coder 2	low	Surface tension is taught classically at secondary level (intermolecular cohesion); no quantum content, so K03's 'quantum physics of... properties of matter' does not fit; no cluster fits.	agree
S5-101	S5	Photon fibres	K10	K07	K10	coder 1	medium	Listed with photon sources, single-photon detectors and vacuum technology; a component technology used to build and run QT (K10), not	agree

Item	Source	Item text	Coder 1	Coder 2	Final	Sided with	Confidence	Reason (adjudicator)	Authors' check
								communication itself (K07).	
S6-011	S6	One-dimensional model or potential well	K02	K01	K02	coder 1	high	Source groups Q11 with wave function and mathematical representations (graphical Schrödinger-equation solutions, wave-function diagrams), so K02, not a fundamental principle (K01).	agree
S6-018	S6	Methodology (e.g., experiments and hypothesis)	K12	K13	K12	coder 1	high	Source lists N1 among nature-of-science aspects (relation between experiment and theory, thought experiments in QP's development); knowledge of how science works, so K12 not K13.	agree

Results. - Of 26 decisions, 18 chose coder 1's cluster and 8 chose coder 2's. 0 chose another cluster, and 0 chose no cluster where neither coder had. - 10 decisions were marked low-confidence. - 8 items changed cluster relative to coder 1. - Retention results that changed (R-retain under both readings, and with S4 and S5 counted as one source): 0. - S6-018, "Methodology (e.g., experiments and hypothesis)", was assigned to K12. Sources listing practices as a separate category: S1. So practices remain a cross-cutting layer (R-crosscut). - Some subjects that appear in several sources are assigned differently by source: the photoelectric effect (S3 → K03, S5 → K01), the double well (S4 → K01, S5 → K02), and standing waves and a particle in a box (S4 → K01, S5 and S6 → K02). The adjudicator decided by each source's own context, for example the key concept under which S4 offers an illustration. These splits fall on the D1–D3 boundaries noted above and change no retention result. - **Authors' check:** done. The authors checked all 26 decisions and changed none of them; the last column records their verdict. - **Neutral re-run of S6-018.** The instructions above single this item out as possibly ambiguous and name its two readings. After a model-run review of the revised manuscript noted this, a fresh run of the same model at the same effort setting decided S6-018 alone, with the same definitions and source texts and with the same instructions except that the paragraph naming the item was removed; the option order was re-randomised (seed 20261027). Decision: K12 Nature, history and philosophy of quantum physics, confidence high; reason: "NOS aspect on scientific method (experiment-theory relation, thought experiments in QP's development): K12 'what it shows about the nature of science', not K13 'activities that learners carry out'." This agrees with the original decision, so practices remain a cross-cutting layer. - Files: `evidence/crosswalk_adjudication.csv` (one row per item) and `evidence/crosswalk_adjudication.json` (summary).

Addendum E — 2026-10-26: post hoc coverage check of further framework, curriculum, and consensus works

Status. Post hoc. The check was run after the domains, the adjudication (Addendum D), and a review of the revised manuscript had been completed, and it does not change the domains. It asks whether the six sources of Section 1, which were chosen before the structured search was complete, missed content that other framework, curriculum, or consensus works list.

Works. The 13 records that the structured search included as framework, curriculum, or consensus works (Supplementary S2, role F; criteria I1, I2, I4, I5) and that are not crosswalk sources: R0001, R0004, R0006, R0007, R0011, R0024, R0033, R0037, R0042, R0102, R0186, R0353, R4135. The four crosswalk sources found by the search and the related framework of Semerikov et al. (2026), with which the manuscript compares the result, are excluded. R0042 reports the same study as crosswalk source S6, and R4135 the same study as R0006; both are counted as separate works.

Reporting fixed before any extraction. (1) For each work: what was read, the number of content items, and the count per domain D1–D9, practice layer, societal dimension, or none. (2) Every item mapped to none is listed verbatim. (3) A topic outside the nine domains is flagged as a candidate the derivation did not see if it appears in at least 3 of the 13 works, at least 2 of them school-level, mirroring R-retain. The flag is reported; the domains are not changed by this check.

Coding. One item per topic, concept, example, or learning standard as the work lists it; rationale, motivation, attitude, and method statements are not content. Items were assigned to the cluster definitions used in the adjudication (Addendum D), calibrated on their member codes, and mapped to domains as in S4_crosswalk_clusters.csv (K01 D1, K02 D2, K03 D3, K05 and K09 D4, K06 D5, K07 D6, K08 D7, K10 D8, K12 D9, K13 practice layer). Citizen-facing societal, ethical, or economic content counts for the societal dimension; K04, K11, K14, and content in no cluster count as none. Curriculum documents, school textbooks, and teachers count as school-level; R0186 covers university and high-school textbooks and is counted as school-level. Extraction and coding were done by a language model (Claude, Anthropic), the model that coded the six sources. A script confirmed that every item occurs verbatim in the saved text of its work. The authors did not re-check the coding item by item.

What was read. Each work was sought in an openly available copy with plain requests. One full text was read (R0024, publisher PDF); the other twelve were coded from their abstracts, because no open copy existed or the host refused automated requests or was unreachable. No request was retried against a host that refused it, and no full text was downloaded by hand. An abstract names fewer topics than a full text, so the counts are lower bounds and the check has low sensitivity to topics outside the domains.

Counts per work.

Work	School-level	Read	Items	D1	D2	D3	D4	D5	D6	D7	D8	D9	practice	societal	none
R0001	yes	abstract	5	3	0	0	0	0	0	0	0	2	0	0	0
R0004	yes	abstract	3	1	0	0	0	0	0	0	0	1	0	0	1
R0006	no	abstract	9	4	1	3	0	0	0	0	0	1	0	0	0
R0007	no	abstract	2	2	0	0	0	0	0	0	0	0	0	0	0
R0011	yes	abstract	3	1	0	1	0	0	0	0	0	1	0	0	0
R0024	yes	full text	45	9	2	12	0	0	1	0	8	11	2	0	0
R0033	no	abstract	0	0	0	0	0	0	0	0	0	0	0	0	0
R0037	yes	abstract	1	0	0	0	0	0	0	0	0	1	0	0	0
R0042	yes	abstract	7	4	0	2	0	0	0	0	1	0	0	0	0
R0102	yes	abstract	2	0	0	1	0	0	0	0	0	1	0	0	0
R0186	yes (mixed)	abstract	1	0	0	1	0	0	0	0	0	0	0	0	0
R0353	yes	abstract	7	6	0	0	0	0	0	0	0	1	0	0	0
R4135	no	abstract	2	0	1	0	0	0	0	0	0	1	0	0	0
Total			87	30	4	20	0	0	1	0	9	20	2	0	1

Items mapped to none (verbatim).

- R0004 (abstract): "general relativity" — relativity: non-quantum topic in the Norwegian curriculum; outside all quantum clusters

Result. - 86 of the 87 items fall within the nine domains or the practice layer (84 in the domains, 2 in the practice layer). Topic "relativity": 1 work(s). - Pre-registered flag: did not fire. - No item fell to D4, D5, or D7 (quantum information, computing, sensing), and none to the societal dimension. The works read concern quantum physics in the curriculum and its teaching; the three technology domains rest on the six sources (manuscript, Section 4.2), and the societal dimension is a design choice, not a result of the derivation. - Teleportation (R0024) is a member code of the communication cluster (K07) in the crosswalk and is counted under D6 here; the placement grid (manuscript, Table 4) treats it under D4, as a use of entanglement as a resource. - Rationale statements that mention society were not extracted as content, among them R0006 "informing and educating society about quantum technologies" and R0037 "the growing relevance of quantum technologies and the need to address quantum misinformation". - File: `S4_coverage_check.csv` (one row per item: work, what was read, verbatim item, locator, cluster, domain, reason).

Supplementary S4, Part B — Domain-derivation rules as frozen, with their dated amendments

Published copy of the project file `crosswalk_rules.md`; SHA-256 of the original: `a9e221fddce3d5143be447f1ab6861c63a496fc1567d7e15e75bca364303d071`. 5 redactions, marked ⟦…⟧, replace internal file names, internal role names, or references to a separate manuscript, and local times are given in UTC; nothing else is changed. R-merge has two clauses: (1) the objectives cannot be distinguished; (2) a cluster has school-level support only at the level of awareness of platforms or devices.

Domain-derivation rules (W2) — FROZEN before any coding

Frozen: 2026-10-24 (time and SHA-256 ⟦recorded in the project log⟧). Any later change must be a dated amendment at the end of this file. Never edit silently.

Sources (six; the extraction must quote each item verbatim with a locator)

- **S1:** CFQT content map. Greinert et al. 2023 (Front. Quantum Sci. Technol. 2:1225733) and CFQT v3.0, Greinert & Müller 2025 (doi 10.2759/8917117). Workforce-oriented.
- **S2:** "Key Concepts for Future QIS Learners" (NSF/NQCO, 2020). Grey literature; verify the official page and record the URL.
- **S3:** Krijtenburg-Lewerissa et al. 2019 (IJSE 41(3):349–366). Delphi study of key topics for secondary school.
- **S4:** Merzel et al. 2024 (EPJ QT 11:27). Core of secondary-level quantum education, multi-stakeholder.
- **S5:** Bitzenbauer et al. 2025 (PRPER 21:020157). Expert Delphi on quantum (technology) topics for school.
- **S6:** Stadermann et al. 2019 (PRPER 15:010130). Topics in the secondary curricula of 15 countries.

School-level sources: S2 (K-12 learners), S3, S4, S5, S6. S1 is NOT school-level.

Procedure

1. **Extraction.** List every topic, concept, competence or content item each source names, verbatim, with a locator (table, page, section) → `crosswalk_extraction.csv` (`source_id, item_id, item_verbatim, source_group, locator`).
2. **Open coding.** Give each item a short content code in the source's own terms. Do NOT use the CFQT domain labels or the R1 domain labels as coding bins.
3. **Second coder.** An author codes all items, or at least 40% of them chosen at random (seed recorded), independently and without seeing the first codes. Report percentage agreement and resolve disagreements by discussion. Until this is done, the domain set is provisional.
4. **Clustering.** Group content codes into candidate domains by shared subject matter.
5. **Apply rules R-retain, R-merge, R-split and R-crosscut, in that order.**
6. **Comparison.** Only after step 5, compare the resulting domains with the CFQT content map, with the nine domains of the R1 manuscript, and with the eight domains of the related published framework (⟦Semerikov et al. (2026)⟧). Record agreements and differences.
7. **Residuals.** Report items that fit no retained domain (e.g., nature of science, history or philosophy of quantum physics), with counts.

Rules

- **R-retain.** A candidate domain is retained if it is supported by items from ≥3 of the 6 sources, of which ≥2 are school-level sources (S2–S6). The CFQT alone cannot create a domain.

- **R-merge.** Two candidate domains are merged if their items within secondary reach (CFQT A1 Awareness / A2 Literacy) cannot be distinguished as learning objectives, OR if one of them has school-level support only at the level of "awareness of platforms/devices".
- **R-split.** A candidate domain is split if its items differ by ≥1 phase in prerequisites (per placement rule P1) AND ≥2 sources treat the parts as separate categories.
- **R-crosscut.** A cluster of practices (programming, simulation, experimentation, collaboration) becomes a domain only if ≥2 sources list it as a separate category. Otherwise it becomes a cross-cutting practice layer noted in every domain.

Expected outcome (a hypothesis recorded in advance, NOT a target)

Eight domains, with "QC hardware & sensors" merged into enabling technologies/platforms and sensors going to sensing. If the rules give a different result, report that result.

Amendments

Amendment 1 — 2026-10-25 ⟦(authors' decision)⟧

- Procedure step 3 originally specified that "an author" codes all items, or at least 40% of them chosen at random, independently. At the authors' decision, the second coding is instead done by a second large language model from a different developer than the first coder (NVIDIA Nemotron 3 Ultra, accessed through Ollama's cloud service, model tag `nemotron-3-ultra:cloud`). It runs independently, in two stages (blind open coding; then assignment of its own codes to the candidate clusters or "none"), and has no access to the first coder's codes, clusters or decisions.
- The sample rule is unchanged: at least 40% of items chosen at random per source with a recorded seed (20261025), plus the pivotal items, unmarked.
- Percentage agreement and Cohen's kappa are reported. Disagreements are listed for adjudication by the authors; until they adjudicate, the domain set stays provisional.
- The effect of the second coder on every retention result is reported (R-retain recomputed under both readings with the second coder's cluster substituted for every sampled item).
- This is an inter-model check, not a human verification of the coding.

Amendment 2 — 2026-10-26 (authors' request). Decided and logged in the project log at ⟦07:00 UTC⟧, before the adjudication ran, together with the reporting plan. Written here at ⟦09:50 UTC⟧, after the run.

- Amendment 1 said that disagreements would be adjudicated by the authors. At the authors' request, they are instead adjudicated by a language model: Claude Opus 5.5 (Anthropic, `claude-opus-5-5`), run as a fresh subagent at the highest reasoning-effort setting. **It is the same model as coder 1**, so a preference for coder 1's assignments is expected. The number of decisions siding with each coder is reported.
- **Blinding:**
- the A/B order is randomised per row (seed 20261026);
- the coders' own code labels are removed;
- the adjudicator works in a folder holding only the packet, the cluster definitions and the six source texts;
- the key linking options to coders is held outside the evidence folder during the run;
- the adjudicator is told not to infer coder identity or consequences.
- The decisions decide only cluster membership. The four rules are then re-applied mechanically under both readings, and with S4 and S5 counted as one source. Every change in a retention result is reported.
- **The authors check every decision before submission.** The number they change is reported. Until the check is done, the manuscript marks it as pending.

Supplementary S4, Part C — Rule-application log (clustering, R-retain, R-merge clauses 1 and 2, R-split, R-crosscut, addenda)

Published copy of the project file `crosswalk_decisions.md`; SHA-256 of the original: `b9a39cdfa7ebfb7fb55926f1bb258b5cb8a9ec45d2926f2d073a0a031e442378`. 3 redactions, marked ⟦...⟧, replace internal file names, internal role names, or references to a separate manuscript, and local times are given in UTC; nothing else is changed.

Crosswalk decision log (W2 / T5) — PROVISIONAL pending the second coder

Written 2026-10-25 (UTC) by the `crosswalk` agent (Claude, model ID `claude-opus-5-5`). Rules: `evidence/crosswalk_rules.md`, sha256 `ffe9bd716970122d7e83d5e0086c33a5392612833059f939c80c6adb88d47fea`, verified unchanged before coding and again before this log was written. Protocol (written before coding): `supplementary/S4_crosswalk_protocol.md`. Data: `crosswalk_extraction.csv` (339 items), `crosswalk_coding.csv` (232 codes), `crosswalk_clusters.csv` (13 candidate clusters). Scripts: `tools/crosswalk/01_extract.py` … `05_outputs.py`.

Read Addendum B (end of file) first: an extraction gap (Merzel et al. Tables 8a–8i) was found after the first rule application; the item set grew from 339 to 400, and the counts and the K09 result below are superseded where Addendum B differs.

Every result below is **provisional** until an author has second-coded the packet (due Thu 29 Oct 2026) and agreement has been computed.

0. Results that contradict the pre-registered hypothesis (read first)

The rules file recorded a hypothesis of eight domains, with "QC hardware & sensors" merged into enabling technologies/platforms and sensors moved to sensing. The rules give a different result:

1. **Societal context is not retained.** Only the CFQT (S1, domain 8 "Valorisation") supplies items about market, business, impact, ethics or public communication. None of the five school-level content lists does. Under R-retain the CFQT cannot create a domain alone, so these 4 items are residual. (The item units were fixed before coding as content lists; the sources' rationale or introduction text, e.g., the NSF introduction on "commerce, governance, privacy, employment, and education", was not eligible. Widening the scope now would be outcome-driven, so it was not done.)
2. **Nature, history and philosophy of quantum physics is retained as a domain** (S3, S4, S5, S6; also under the endorsement reading), although the rules file lists "nature of science, history or philosophy" as an example of a residual.
3. **Atoms, light and matter** and **Quantum information and qubits** come out as domains of their own; R1's "Physical foundations" is divided.
4. **Qubit hardware platforms** do not form a domain and do not merge anywhere: they fail R-retain and are reported as residuals (§6), not absorbed.
5. **Communication and sensing are fragile:** they are retained only under the literal reading of "support" (§2).

Primary outcome: **nine domains plus a cross-cutting practice layer**, with 17 residual items. Endorsement reading: seven domains.

1. Candidate clusters (step 3)

Items per source, literal reading (endorsement reading in brackets). S1 = CFQT (workforce); S2–S6 school-level.

ID	Candidate cluster	codes	S1	S2	S3	S4	S5	S6	items
K01	Quantum principles and phenomena	48	1 (1)	10 (10)	22 (11)	27 (24)	33 (7)	6 (6)	99 (59)
K02	Mathematical representations	24	1 (1)	4 (4)	6 (3)	8 (8)	12 (1)	3 (3)	34 (20)
K03	Atoms, light and matter	46	4 (4)	0	24 (15)	7 (7)	27 (1)	6 (6)	68 (33)
K04	Particle and nuclear physics	4	0	0	3 (1)	0	1 (0)	0	4 (1)
K05	Quantum information and qubits	10	0	9 (9)	1 (1)	2 (2)	1 (0)	0	13 (12)
K06	Quantum computing and simulation	11	5 (5)	4 (4)	1 (1)	2 (1)	1 (0)	0	13 (11)
K07	Quantum communication and cryptography	12	6 (6)	3 (3)	4 (0)	1 (0)	4 (0)	0	18 (9)
K08	Quantum sensing and metrology	17	7 (7)	5 (5)	2 (0)	0	6 (0)	0	20 (12)
K09	Qubit hardware platforms	9	8 (8)	1 (1)	0	0	0	0	9 (9)
K10	Quantum devices and enabling technologies	28	4 (4)	0	17 (4)	0	12 (0)	1 (1)	34 (9)
K11	Valorisation and societal impact	4	4 (4)	0	0	0	0	0	4 (4)
K12	Nature, history and philosophy of QP	15	0	0	4 (1)	2 (2)	6 (1)	7 (7)	19 (11)
K13	Practices (programming, experimentation, building)	4	2 (2)	0	0	0	2 (0)	0	4 (2)

Boundary placements made at clustering, with the rule used: an item goes to sensing when the source links it to precision measurement or timing (GPS: S2-004 "depends on the extreme precision of atomic clocks"; S3-074 by the same code); medical and laboratory instruments whose working principle the item names but not a measurement-precision claim go to devices (MRI, NMR, CT, PET, STM, SEM, AFM). Sensor devices named as sensors (S1 6.2–6.4; S5 "Superconducting sensors", "Spin-qubit based sensors", "Atom interferometers") go to sensing; physical systems named as qubits or qubit control go to K09. Neither boundary changes any pass/fail result (checked by `04_pivotal.py`), except the K09 sensitivity in §5.

2. R-retain (≥ 3 of 6 sources, ≥ 2 school-level; the CFQT alone cannot create a domain)

Which reading do the frozen rules specify? The rule says a domain "is supported by items from ≥3 of the 6 sources". It counts sources whose *items* support the domain and says nothing about the sources' own ratings or inclusion thresholds. The literal reading (an extracted item coded into the domain counts) is the reading closest to the text, and it was pre-registered as primary in protocol §2 before any item was coded. The rule is nevertheless **ambiguous**: "support" is undefined, and the extraction includes topics that a source rated and rejected (S5 Tables IV–V: no Physical-Foundations or Quantum-Technologies topic met both inclusion criteria, and all have Cns < 0.75 and inclusion below 75 %; the authors write that advanced QT topics "were widely deemed unsuitable for the secondary school"). The primary result below uses the literal reading; the endorsement outcome is reported beside it. **Flag for the authors:** if they prefer the endorsement reading, it must be added as a dated amendment to `crosswalk_rules.md`, and the domain count becomes seven.

Cluster	Literal: sources (school)	Result	Endorsed: sources (school)	Result
K01	6 (5)	retain	6 (5)	retain
K02	6 (5)	retain	6 (5)	retain
K03	5 (4)	retain	5 (4)	retain
K04	2 (2)	**fail**	1 (1)	fail
K05	4 (4)	retain	3 (3)	retain
K06	5 (4)	retain	4 (3)	retain
K07	5 (4)	retain	2 (1): S1, S2	**fail → fragile**
K08	4 (3)	retain	2 (1): S1, S2	**fail → fragile**
K09	2 (1): S1, S2	**fail**	2 (1)	fail
K10	4 (3)	retain	3 (2): S1, S3, S6	retain
K11	1 (0): S1 only	**fail**	1 (0)	fail
K12	4 (4)	retain	4 (4)	retain
K13	2 (1)	(practices: decided by R-crosscut)	1 (0)	—

K07 and K08 fall to S1 and S2 under the endorsement reading because every S3, S4 and S5 item in them is below that source's own cut-off (S3: atomic clock 6/12, GPS 5/12, quantum cryptography 7/12, teleportation 6/12, BB84 3/12; S4: quantum cryptography offered as a sparse illustration; S5: all Table V topics < 75 % inclusion, e.g., BB84 34 %, atom interferometers 29 %, Definition of SI units 51 %). This matches Bitzenbauer et al.'s own conclusion. The manuscript must not describe communication or sensing as consensus domains.

3. R-merge (applied to retained clusters)

Clause 1 test: write the A1/A2 objective each cluster implies (CFQT A1 "Reproduce basic quantum concepts & terminology"; A2 "describing and relating quantum concepts, including elements of mathematical formalism", as quoted by S5 §VII.B) and ask whether the two can be told apart. Clause 2 test: is a cluster's school-level support only "awareness of platforms/devices"?

Pair	Objective implied by cluster A (items)	Objective implied by cluster B (items)	Decision
K01–K02	describe and relate phenomena: superposition, interference, measurement outcomes, uncertainty (S4-001 "Superposition"; S6-005 "Wave-particle duality or complementarity")	represent states and compute outcome probabilities: "Quantum states are represented by directions or vectors in an abstract space" (S2-006); "Calculation of detection	distinguishable (representation vs phenomenon; S4 and S6 class them separately; S5 treats the mathematical level as a separate instructional level) → **no merge**

Pair	Objective implied by cluster A (items)	Objective implied by cluster B (items)	Decision
		probability, Born rule" (S6-017); Dirac notation, matrices (S5-003/004)	
K01–K03	as above	explain spectra, emission and material properties from quantised levels and bands (S6-003 "Discrete energy levels (line spectra)"; S3-036 "Spectral lines"; S5-077 "Band structure")	distinguishable (principle vs its application to atoms and solids) → **no merge**
K01–K05 (most contestable)	"Spin-1/2 particle (two level system)" (S5-010); "Two-level system" (S4-029): describe measurement outcomes of a two-state system	"Unlike a classical bit, each qubit can represent information in a superposition" (S2-015); "a set of n qubits can exist in a superposition of all of these states" (S2-016); "The quantum bit, or qubit, is the fundamental unit of quantum information" (S2-014)	the information objectives (bit–qubit contrast, encoding, 2^n scaling) occur in no K01 item → **no merge**. Alternative recorded and not adopted: Merzel et al. §5.3.2/§6.2 note that "Qubit" and "Two level systems" were offered both as key concepts and as illustrations ("indeterminacy"), and the CFQT lists "Two-state systems (e.g. spin-1/2, polarisation), qubits" together in 1.1. If the second coder clusters "qubit" with "two-level system", K05 fails R-retain (pivotal, §5).
K05–K06	qubit as unit of information (above)	"will solve certain complex computational problems more efficiently … no speed advantage for many other types of computations" (S2-028); "Quantum gates (esp. H. Z. X. CX)" (S5-089); error correction (S2-029)	distinguishable → **no merge** (Semerikov et al. (2026) place "Qubit concept" inside Quantum computing; the crosswalk does not)
K06–K07–K08	computing (above)	communication: eavesdropper detection (S2-032), teleportation (S2-031); sensing: "detect and measure physical properties with the highest precision allowed by quantum mechanics" (S2-033)	distinguishable; S2 itself presents them as three areas → **no merge**
K03–K10	energy levels, bands, materials (above)	explain how devices work and name the technologies QT require: "Technical applications (e.g., scanning electron microscope SEM, semiconductors, LED, laser)" (S6-007); "Solar cells", "Lasers", "LEDs" (S3)	distinguishable (device function vs underlying physics; S3 and S6 separate applications from concepts) → **no merge**
K08–K10	precision measurement, quantum vs classical sensors, SI units (S2-036; S5-098)	device function (above)	distinguishable → **no merge**
K01–K12	principles (above)	discuss interpretations, history, models, tentativeness (S6-010 "Philosophical consequences or interpretations"; S6-019–023)	distinguishable → **no merge**

Clause 2: K07 has conceptual school items (S2-032 eavesdropper detection; S3-083 "Bennet–Brassard protocol"; S5-092 "Comparison to current cryptography technologies"); K08 has conceptual school items (S2-033, S2-034, S2-036; S5-098 "Quantum sensors versus classical sensors"); K10 has curriculum learning outcomes (S6-007, present in 13 of 15 countries). None has school-level support only at the level of awareness of platforms/devices → **no merge under clause 2**. K09, the cluster for which clause 2 was written, never reaches R-merge because it fails R-retain.

4. R-split (both conditions needed: ≥ 1 phase difference in prerequisites per P1 AND ≥ 2 sources separate the parts)

P1 evidence available now: Stadermann et al. 2019 (QP mostly in final-year advanced courses, ages 17–19; only Bavaria teaches it in a compulsory course at 15–16; entanglement in 2 of 15 countries' documents; discrete energy levels in 15 of 15; no materials or band-structure item); Lehka 2021 Appendix Γ as summarised in the project's

internal notes (not re-read for this log): Ukrainian standard grade 10 quantum optics/photoeffect, grade 11 Bohr model, lasers, de Broglie; uncertainty and tunnelling only at profile level; probability and complex numbers only in grade-10 advanced mathematics; no matrices. `evidence/evidence_matrix.csv` (T2) does not yet exist, so **every "not established" below is provisional pending T2/T6.**

Candidate split	≥ 2 sources separate the parts?	≥ 1 phase prerequisite difference (P1)?	Decision
K01: entanglement/non-locality vs other principles	yes: S3 Table 5 "nonlocality"; S4 categories "Entanglement", "Non-locality"; S5 Table I "Entanglement and nonlocality" (CFQT-structured)	not established: entanglement's prerequisites (superposition, probabilistic measurement, correlation) are the same probability-based prerequisites as the rest of K01; the curricula differ in *presence* (2/15 vs 15/15) but P1 concerns prerequisites	**no split** (provisional)
K03: atoms vs materials/solid state	yes: S3 Table 5 "atoms" vs "materials"; S5 Table I "Atoms and atomic physics" vs "Solid state physics" / "Material science and nanostructures" (CFQT-structured)	not established: atomic levels are core curriculum (15/15, ages 17–19; Ukrainian grade 11); no P1 source reports where band structure sits	**no split** (provisional)
K10: first-generation devices vs QT enabling technologies	no: only S5 separates them (Tables IV vs V, CFQT-structured); S1 has no first-generation-device category; S3 groups all devices as "applications"; S6 has one item	—	**no split**
K12: history vs philosophy/interpretation vs NOS	yes: S6 (Table III philosophical theme vs Table I NOS/history aspects); S3 (a "history" category)	no: NOS aspects appear in ≥ 12 of 15 countries' curricula in all QP courses; no curricular prerequisite separates them	**no split**

5. R-crosscut and pivotal items

K13 contains "Computers and software" (S1-011), "Quantum programming tools …" (S1-022), "Measuring h" (S5-037) and "Microelectronics (Arduino. Raspberry)" (S5-103). Sources that list practices as a separate category: S1 (proficiency area II "QT hardware (HW) & software (SW) engineering"). S6 lists "Methodology (e.g., experiments and hypothesis)" inside its table of *nature-of-science* aspects; it was coded as understanding methodology (Table I example: "relation between experiment and theory", "thought experiments") and clustered in K12, so S6 is not counted as listing practices. One source < 2 → **practices become a cross-cutting practice layer noted in every domain**, not a domain. (If N1 were coded as a practice, S6 would arguably make two sources, and R-crosscut would yield a practice domain; this item is in the purposive add-on, §8.)

Pivotal items (recoding one would flip a result; from `04_pivotal.py`):

Result	Reading	Pivotal item(s)
K05 retained	both	code "qubit" (S2-014, S4-015, S4-040, S5-054); endorsed also S3-059 "Quantum information"
K06 retained	endorsed	S3-063 "Quantum computers" (8/12, exactly at cut-off) or S4-016 "Quantum computing" (jointly)
K10 retained	endorsed	S6-007 "Technical applications"
K12 retained	endorsed	code "Schrödinger's cat" (S3-046, S5-033)
K09 fails	literal	S2-024 (NSF 6b); S5-100 "Entangled photon sources", S5-093, S5-094 (one S5 item coded as a platform would give K09 three sources)
practice layer	—	S6-018 "Methodology (e.g., experiments and hypothesis)"

6. Resulting domains (primary = literal reading), definitions and support

Domain	Definition (one sentence)	Items (endorsed)	Sources literal / endorsed
D1 Quantum principles and phenomena	Superposition, interference, measurement and probability, uncertainty and complementarity, wave–particle duality, quantisation, spin, entanglement and decoherence, with the experiments that display them.	99 (59)	6 / 6
D2 Mathematical representations	State vectors, wave functions, amplitudes and the Born rule, operators and model systems such as the potential well.	34 (20)	6 / 6
D3 Atoms, light and matter	Quantum accounts of atomic structure and spectra, light–matter interaction, and the properties of solids and materials.	68 (33)	5 / 5
D4 Quantum information and qubits	The qubit as a unit of information in a two-state system, its contrast with the bit, multi-qubit states, and what quantum information science is.	13 (12)	4 / 3
D5 Quantum computing and simulation	What quantum computers and simulators do, gates and algorithms, speed-up and its limits, error correction.	13 (11)	5 / 4
D6 Quantum communication and cryptography *(fragile)*	Quantum cryptography and key distribution, teleportation, random-number generation and networks.	18 (9)	5 / **2 (fails)**
D7 Quantum sensing and metrology *(fragile)*	Precision measurement with quantum states: sensors, atomic clocks and timing, quantum limits, SI units.	20 (12)	4 / **2 (fails)**
D8 Quantum devices and enabling technologies	Devices that work because of quantum physics (lasers, LEDs, solar cells, detectors, microscopes, medical scanners) and the techniques used to build quantum technologies.	34 (9)	4 / 3
D9 Nature, history and philosophy of quantum physics	Historical development, interpretations, the role of models and other nature-of-science aspects of quantum physics.	19 (11)	4 / 4
Practice layer (cross-cutting)	Programming, simulation, experimentation and building, noted in every domain.	4 (2)	—

7. Comparison (done only after §§2–5)

CFQT content map (8 domains). D1 ≈ CFQT 1.1; D2 ≈ 1.2; D3 ≈ domain 2 (2.1–2.4); D4 has no CFQT domain (the CFQT places qubits in 1.1 and information theory in 1.2); D5 ≈ domain 5 (5.3 programming tools → practice layer); D6 ≈ domain 7; D7 ≈ domain 6; D8 ≈ domain 3 (3.1–3.4; 3.5 → practice layer) plus first-generation devices that the CFQT does not list; **CFQT 4 "Quantum hardware" → no domain (residual)**; **CFQT 8 "Valorisation" → no domain (residual)**; D9 has no CFQT counterpart.

R1 manuscript (9 domains). Physical foundations → D1 + D3 (and D4 for qubits). Mathematical foundations → D2. Supporting technologies → D8 (widened to first-generation devices). **QC hardware & sensors → not retained:** sensors → D7; qubit platforms → residual. Quantum computing → D5 (qubit concept moved to D4). Quantum

sensing → D7 (fragile). Quantum communication → D6 (fragile). Practical skills → cross-cutting layer. **Societal context → not retained (residual).** New: D3 as a separate domain, D4, D9.

Semerikov et al. (2026) (8 domains, Table 3, p. 7). Physical foundations (superposition, entanglement, measurement, wave–particle duality) → D1 (its content is CFQT 1.1, not CFQT domain 2, so the same label means different things in the two frameworks); Mathematical foundations → D2; Quantum computing (incl. "Qubit concept") → D5 + D4; Quantum sensing → D7; Quantum communication → D6; Supporting technologies (cryogenics, photonics, control systems, infrastructure) → D8; **Applications and impact (industry applications, societal benefits, ethics, careers) → no domain**; Practical skills → practice layer. Semerikov et al. (2026) have no atoms/matter domain, no information domain and no nature-of-science domain; like the crosswalk, they have no separate hardware domain.

Why (or whether) hardware/sensors and sensing are separate — answer to R1.2a. They should not be separate, and in the derivation they are not. (a) Sensor devices code to sensing by subject matter: the CFQT's sensor subdomains (6.2–6.4) and S5's "Superconducting sensors", "Spin-qubit based sensors" and "Atom interferometers" all sit in D7. (b) Qubit hardware platforms (CFQT 4.1–4.8) form a candidate cluster that fails R-retain: apart from the CFQT, only one NSF fundamental (6b, "Some types of qubits are inherently isolated, whereas others require carefully-engineered materials …") addresses them, and no school-level Delphi or curriculum source names a qubit platform. So no "hardware" domain survives, and the R1 split had no support beyond the CFQT's workforce structure (CFQT v3.0 itself keeps "4 Quantum hardware" and "6 Quantum sensors and imaging systems" apart). Observation only: the R1 label "QC hardware & sensors" closely mirrors S5's round-1 category "Hardware for quantum computers and sensors" (9 mentions), which S5's authors assigned "using the CFQT". An internal planning note states that the split was inherited from the CFQT v1.0 content map; this was not verified against the Zenodo record. ⟦The manuscript⟧ may present qubit platforms as awareness-level context inside D8, but must say that this placement is an authors' decision and **not derived from the crosswalk**.

8. Residuals, second coding, limitations

Residuals (items in failed clusters; fit test = the §5 clustering criterion, so none is re-homed): 17 of 339 items (5.0 %). - K09 Qubit hardware platforms: 9 (S1 4.1–4.8: 8; S2 6b: 1). - K04 Particle and nuclear physics: 4 (S3 "Subatomic particles", "Standard model", "Radioactive decay"; S5 "Pair production"). - K11 Valorisation and societal impact: 4 (S1 8.1–8.4). - Under the endorsement reading, K07 (18 items) and K08 (20 items) would also be residual (55 items in all).

Second coder. Packet `crosswalk_second_coding_packet.csv`: a random 40 % stratified by source (seed 20261025) plus an unmarked purposive add-on of the pivotal items in §5 that the random draw missed. Agreement on the random sample and on the pivotal items is reported separately (instructions in `SECOND_CODER_INSTRUCTIONS.md`).

What a reviewer could criticise. 1. One model did extraction, coding and clustering, and had seen the source structures; the second coder is the only check. 2. Clustering granularity is a judgement. Coarser clusters would merge D1/D3 (as Semerikov et al. (2026) and R1 did); finer ones would create more residuals. 3. "Support" is undefined in the frozen rules; two domains depend on the reading. 4. Several retentions rest on one or two items (§5). 5. Source independence is limited. S4 reports the first round of the same QCI Delphi study whose three rounds S5 reports (S5 §IV.B: "previously reported by Merzel et al."). S4 respondents were shown the CFQT, S5 structured its categories with the CFQT, and S5's authors include the CFQT authors (Greinert, Müller) and the first authors of S3, S4 and S6. Treating S4 and S5 as one source changes no R-retain result under either reading (checked by hand from §2), but it makes K12 (endorsed) depend on S3's "Schrödinger's cat". 6. S2 is a non-consensus workshop document, and S1 is workforce-oriented. 7. Item units differ in grain: one CFQT subdomain is counted like one S3 topic. Counts are therefore read as source presence, not weight. 8. S4 illustrations come only from the article text; the image tables 7a–9b (104 illustrations) were not extracted. 9. The P1 judgements in R-split rest on two curriculum sources and are provisional. 10. English-language sources only.

Addendum B — 2026-10-25 (UTC): extraction gap found after the first rule application

What happened. After §§1–8 were written, the authors' review noted that Merzel et al.'s image tables had not been extracted. The publisher PDF could not be retrieved (anti-bot challenge), so the PDF pages could not be rendered; instead the publisher's online table pages and full-size table images were retrieved (online tables 7–27). Tables 7a–7h list KC categories per prominent illustration (no new illustrations); Tables 9a–9b are consensus tables (HTML); **Tables 8a–8i list the illustrations offered for the nine most prominent KC categories.** All nine images were read and transcribed (118 cells; Merzel et al.'s Table 6 gives 127 per-category distinct illustrations for these categories: 8a

20 vs 21, 8c 14 vs 20, 8e 11 vs 12, 8f 12 vs 13; the images show complete tables, so the difference is recorded as an inconsistency in the source, not filled in). Illustrations for the other ten KC categories are not tabulated by the source. PDF page numbers are not available; locators give table, online table number, row and column.

Extraction. 61 distinct illustrations not already extracted were added as S4-050 to S4-110 (variants of existing items were mapped to them, e.g., "BB84" and "Ekert 91" to S4-049; "Spin states" to S4-042). Their verbatim check is against our own transcription only (`image_transcription`), not an independent match. By Merzel et al.'s own cut-off (at least 6 mentions and at least 4 KC categories), **none of the 61 is endorsed.** Items: 400 (S4: 110).

Coding and clustering. Codes were assigned under protocol §4 (22 new codes; 254 in all). "Bernstein 1967" (S4-099) cannot be coded from its wording and forms an unclassifiable cluster K14 that cannot be retained. Two judgements decide a result and are flagged: S4-097 "Coherent control" and S4-098 "NV-center" were coded as qubit control / qubit platform (K09). Both were offered once, by physicists, for KC "Qubit". "NV-center" names neither a qubit nor a sensor, and the CFQT lists NV centres under both 4.2 and 6.2, so the source's KC context was used to resolve it; the alternatives are K01 (two-state dynamics) and K08 (sensor).

R-retain, 400 items (sources literal / endorsed; items literal (endorsed)):

Cluster	Items	Literal	Endorsed	Change from §2
K01 principles	130 (59)	6 retain	6 retain	S4 27→58 items
K02 mathematical representations	44 (20)	6 retain	6 retain	S4 8→18
K03 atoms, light and matter	75 (33)	5 retain	5 retain	S4 7→14
K04 particle/nuclear	4 (1)	2 fail	1 fail	none
K05 quantum information and qubits	13 (12)	4 retain	3 retain	none
K06 quantum computing	16 (11)	5 retain	4 retain	S4 2→5 (Hadamard gate, Quantum circuit, Bernstein-Vazirani alg.)
K07 communication	19 (9)	5 retain	**2 fail**	S4 +Teleportation; **dispute not resolved**
K08 sensing	20 (12)	4 retain	**2 fail**	none: no sensing illustration in 8a–8i
K09 qubit hardware platforms	11 (9)	**3 retain** (S1, S2, S4)	2 fail	**changed: was 2 fail**
K10 devices and enabling technologies	34 (9)	4 retain	3 retain	none
K11 valorisation/societal	4 (4)	1 fail	1 fail	none
K12 nature/history/philosophy	24 (11)	4 retain	4 retain	S4 2→7
K13 practices	5 (2)	3 (R-crosscut)	1	S4 +Quantum games
K14 unclassifiable	1 (0)	1 fail	0 fail	new

The reading dispute for D6/D7 does not disappear. The new communication illustrations (Teleportation, offered by two groups; BB84 and Ekert 91, each by one) and the absence of any sensing illustration leave both domains at two sources under the endorsement reading. (⟦The authors have⟧ since confirmed the literal reading as primary and the endorsement reading as the sensitivity check.)

K09 — two branches; no separate hardware domain in either. - **Branch A (literal reading, current coding):** K09 passes R-retain on S1 (8 subdomains), S2 (6b) and S4 (S4-097, S4-098). Its school-level support is one NSF fundamental and two single-mention illustrations, i.e. awareness of platforms only, so **R-merge clause 2 applies**. Partner criterion (stated before choosing, the same test as clause 1): the retained domain whose A1/A2 objective the school-level platform items serve. NSF Key Concept 4 defines the qubit as "encoded in a physical system, such as polarization states of light, energy states of an atom, or spin states of an electron"; NSF 6b concerns "types of qubits"; S4 offered both illustrations for the KC "Qubit". → **merged into D4 (Quantum information and qubits), which then includes awareness of physical qubit platforms.** Alternative considered and not adopted: D8 (devices and enabling technologies), which would rest on a subject-matter resemblance ("hardware is a kind of device"), not on how the school-level sources use the items. - **Branch B (endorsement reading, or either S4-097/S4-098 recoded):** K09 fails R-retain (S1, S2) and its 11

items are residual. - Wording consequence for the manuscript and the response letter: "the hardware domain fails R-retain" is true only in Branch B; under the primary reading it is an R-merge (clause 2) outcome. Sensor devices go to D7 by subject matter in both branches.

Pivotal items (400-item set). Literal: K05 on the code "qubit" (S2-014, S4-015, S4-040, S5-054); K09 on S2-024, S4-097, S4-098 (each). Endorsed: K05 on S3-059 and "qubit"; K10 on S6-007; K12 on "Schrödinger's cat" (S3-046, S5-033); K06 on S3-063 or S4-016 jointly. Practice layer: S6-018. S5-093/094/100 are no longer pivotal.

Independence check repeated. Treating S4 and S5 as one source changes no R-retain result under either reading (K09 literal: S1, S2, S4/S5 = 3).

Resulting domains (primary, Branch A): D1 130 items; D2 44; D3 75; D4 24 (13 + 11 merged platform items); D5 16; D6 19 (fragile); D7 20 (fragile); D8 34; D9 24; practice layer 5. **Residuals: 9 of 400** (K04 4, K11 4, K14 1). Endorsement reading: 7 domains; residuals 59 of 400 (K04 4, K11 4, K14 1, K07 19, K08 20, K09 11).

Comparison with the R1 domains and the CFQT v3.0 content map (claim check). The R1-round response draft states that "the R1 domains were taken over, with one addition, from the CFQT content map". Read against the R1 domain definitions (R1 manuscript, Table 2 and §4.2) and CFQT v3.0 (the version the R1 manuscript cites as v3), the correspondence is:

CFQT v3.0 domain	R1 domain(s)
1 Concepts and foundations	**two** R1 domains: "Physical foundations" (superposition, interference, entanglement, measurement = CFQT 1.1) and "Mathematical foundations" (= CFQT 1.2)
2 Physical foundations of quantum technologies (atomic, optics, solid state, many-body)	**none** — R1 reuses the label "Physical foundations" for CFQT 1.1 content
3 Enabling technologies and techniques	Supporting technologies
4 Quantum hardware	"QC hardware & sensors" (which also adds sensor principles that the CFQT keeps in domain 6)
5 Quantum computing and simulation	Quantum computing
6 Quantum sensors and imaging systems	Quantum sensing (and partly "QC hardware & sensors")
7 Quantum communication and networks	Quantum communication
8 Valorisation (8.3 impact and responsibility incl. ethics; 8.4 education and communication)	partly Societal context, although R1 says this row "supplies the societal dimension absent from the CFQT"
—	Practical skills (no CFQT content domain; CFQT 3.5/5.3 subdomains and proficiency area II)

The claim is therefore **approximate at best**: one CFQT domain is split into two R1 domains, one CFQT domain has no R1 counterpart despite the shared label, sensors appear in two R1 domains, Practical skills has no CFQT domain, and Societal context overlaps CFQT domain 8 although R1 calls it absent from the CFQT. ⟦The manuscript⟧ should not repeat "with one addition".

Origin of the R1 label "QC hardware & sensors" (checked 2026-10-25, UTC). Lehka (2021, p. 28) renders the CFQT v1.0 (2021) structure as seven knowledge fields and one practical-skills section: Concepts of quantum physics; Physical foundations of quantum technologies; Supporting technologies; **Hardware for quantum computers and sensors**; Quantum computing and simulation; Quantum sensors and metrology; Quantum communication; Practical skills and key competences. Lehka's own competency model (p. 54) has eight groups: physical foundations; mathematical foundations of quantum informatics; enabling technologies; hardware for quantum computers and sensors; quantum computing and simulation; quantum sensors and metrology; quantum communication; practical skills and general competences. Bitzenbauer et al. (2025, Table I) use the same category "Hardware for quantum computers and sensors", assigned with the CFQT. **The R1 nine domains equal Lehka's eight groups plus Societal context.** So "taken over, with one addition" holds approximately against CFQT v1.0 as rendered by Lehka (2021), with v1.0's concepts/physical-foundations pair re-cut into physical vs mathematical foundations, but not against CFQT v3.0, the version the R1 manuscript cites, where the domain is "Quantum hardware" and sensors sit only in "Quantum sensors and imaging systems". CFQT v1.0 itself (Zenodo) was not opened; the rendering is Lehka's (in Ukrainian; translation by the agent).

Addendum C — 2026-10-25 (UTC): inter-model second coding (Amendment 1)

Second coder: NVIDIA Nemotron 3 Ultra (`nemotron-3-ultra:cloud`, via Ollama 0.30.6), two stages, blind to the first coder's codes; prompts, settings and raw records in `supplementary/S4_crosswalk_protocol.md` Addendum C and `evidence/second_coder_llm/`. **This is an inter-model check, not a human verification.**

- Agreement on cluster assignment: random sample 138/162 = 85.2 % (Cohen's kappa 0.826); pivotal items 11/14 = 78.6 % (kappa 0.731); all 170 packet items 84.7 % (kappa 0.822).
- 26 disagreements, listed in `evidence/crosswalk_disagreements.csv` for the authors (empty `author_decision` column). The most frequent were mathematical representations → principles (6) and atoms, light and matter → principles (5).
- **R-retain sensitivity:** with the second model's cluster substituted for all 170 sampled items, **no retention result changes under either reading.** Source sets change only for D7 and D8 (+S4, literal), the practice layer (+S6) and D9 (endorsed: loses S3 and is then supported by S4, S5 and S6, still three sources).
- **K09 (qubit platforms), Branch A** survives (S1, S2, S4) but then rests on one single-mention illustration, "NV-center", because the second model placed "Coherent control" (S4-097) with devices.
- **R-crosscut caveat (needs adjudication):** the second model placed S6-018 "Methodology (e.g., experiments and hypothesis)" with practices. If the authors adopt that, a second source arguably lists practices as a separate category and R-crosscut would turn the practice layer into a domain (ten domains on the literal reading).
- The second model was internally inconsistent on "Schrödinger's cat" (same label, two clusters).
- **Conditions on the agreement figures.** The stage-2 cluster definitions were written by the first coder (from the labels and the rules file only), and the clusters are the first coder's candidate clusters, so the figures measure agreement on assignment to that cluster set, not an independent clustering. The disagreements concentrate on the D1/D2/D3 boundaries, which supports the granularity limitation (§8, item 2) and changes no result.
- **Combined sensitivity (S4 and S5 counted as one source):** literal reading — no change with either coder, except that the practices cluster would fail R-retain (no consequence: practices are a layer by R-crosscut); endorsement reading with the second model's clusters — **D9 (nature, history and philosophy) would fail** (sources S4/S5, S6). The primary result is unaffected; add this to the limitations.

Until the authors have adjudicated the disagreements, the domain set stays provisional.

Supplementary S4 — Run records of the model adjudication

Adjudication of the 26 disagreements: prompt and run record

- **Requested by:** the corresponding author, in the session on 2026-10-26: "Use Opus 5.5 with Max efforts for adjudication". The authors will check every decision before submission.
- **Model:** Claude Opus 5.5 (`claude-opus-5-5`, Anthropic). It ran as a fresh Claude Code subagent with reasoning effort `max` and no access to the main model session's context.
- **Date:** 2026-10-26.
- **Relation to the coders:** coder 1 was the same model (`claude-opus-5-5`); coder 2 was Nemotron 3 Ultra (NVIDIA, via Ollama cloud).
- **Blinding measures:**
- The A/B order is randomised per row (seed 20261026; the same order as the authors' packet).
- The coders' own code labels are removed from the options, because their capitalisation and wording could identify the coder. The authors' packet (`evidence/adjudication/ADJUDICATION_PACKET.csv`) keeps them.
- The unblinding key was moved out of the project's evidence folder for the duration of the run.
- The adjudicator was told to read only this folder and not to infer coder identity or consequences.
- **Inputs:** `INSTRUCTIONS_FOR_MODEL.md`, `PACKET_FOR_MODEL.csv`, `cluster_definitions.md`, `sources/`.
- **Output:** `DECISIONS_MODEL.csv`, `progress.log`, `SUMMARY.md`.
- **Prompt given to the subagent:** see below, verbatim.

You are the adjudicator for a coding exercise in an academic study. Your working folder is: /adjudication_run/

Read INSTRUCTIONS_FOR_MODEL.md in that folder first and follow it exactly. In short: - decide all 26 rows of PACKET_FOR_MODEL.csv, using only cluster_definitions.md and the source texts in sources/; - write DECISIONS_MODEL.csv atomically after every row and append to progress.log; - finish with SUMMARY.md.

Rules: - Work only inside this folder. Do not open, list or search any file or directory outside it, and do not use the web. - Do not try to infer which coder gave which option, or what a decision would change. - Quote paths in shell commands, because they contain spaces, '&' and parentheses. Use python3 -I for any script. - Take the time each row needs. Read each item in its source context before deciding.

When done, reply with: - the number of rows decided; - the counts of each decision type; - the path of DECISIONS_MODEL.csv.

Do not include anything else.

Interruption and resumption (2026-10-26). The run stopped when the account's usage limit was reached. At that point the adjudicator had read every item but saved no decision. After the account holder logged in again, it was resumed in the same context with this message (verbatim):

> Resume the adjudication: the usage limit that stopped you has been cleared. Nothing has been saved to disk yet. Write your decisions now, following INSTRUCTIONS_FOR_MODEL.md: after each row, rewrite DECISIONS_MODEL.csv atomically and append a line to progress.log. If you consult the advisor on borderline rows, save the rows you are sure of first. Then write SUMMARY.md and reply in the format requested in your original prompt. All other rules are unchanged: work only inside adjudication_run/.

The adjudicator finished at 09:19 UTC, with all 26 rows decided. It could not create SUMMARY.md because subagents cannot write report files. The main model session saved that file verbatim from the adjudicator's final report.

The subagent also had access to an "advisor" tool. That tool sends the subagent's own transcript to a stronger reviewer model, so it adds no information from outside the adjudication folder. The subagent's transcript shows 3 advisor calls; the advice itself is redacted in the transcript and is not recorded.

Adjudication of the 26 disagreements: the model's summary

(Written by the main model session from the adjudicator's final report, verbatim, because the subagent could not create report files; 2026-10-26.)

All 26 rows of `PACKET_FOR_MODEL.csv` are decided. The decisions are in `DECISIONS_MODEL.csv`, which has the packet's columns plus `confidence`.

Decision counts

Decision	Rows
A	12
B	14
other	0
none	0

Row 1 is counted as A. Its option A is itself "none (no cluster fits)", so choosing A there means that no cluster fits.

Confidence levels: 10 high, 6 medium and 10 low.

Low-confidence rows

Row	Source	Item	Decision	Why confidence is low
1	S5	Surface tension	A (none)	K03 is the matter and materials cluster, but surface tension has no quantum content as taught at secondary level.
3	S4	Analogy of standing waves and a particle in a box	B (K01)	S4 offers it as an illustration of Quantization. The particle-in-a-box model can also be read as a representation of states (K02).
4	S5	Electrons	B (K03)	A bare term. K04 ("subatomic particles") fits it literally; K03 rests on its neighbours in the sequence (spin, Franck-Hertz experiment, atomic models).
11	S3	Heat radiation	B (K03)	Light emitted by matter (K03), but also the phenomenon in which energy quantisation first showed itself (K01).
12	S3	Single molecule microscopy	B (K08)	A measurement and imaging technique (K08), but also readable as a device

Row	Source	Item	Decision	Why confidence is low
				whose operation relies on quantum physics (K10).
15	S3	Stationary states	B (K02)	Defined by how a state changes over time (K02), but taught at secondary level together with energy quantisation (K01).
17	S5	Quantum harmonic oscillator	A (K02)	A model system (K02), but it also exemplifies quantised energy (K01). Its neighbours in the sequence are mixed.
18	S3	Schrödinger's cat	A (K12)	Mainly a vehicle for discussing interpretation (K12), but S3 lists it in its Examples column beside experiments (K01).
21	S4	Double-well	A (K01)	S4 offers it for four principle-type key concepts, but it is in itself a model potential described by wave functions (K02).
25	S3	Free vs. localised particle	A (K01)	The source gives no description. It is a conceptual principle (K01), but could also be read as a wave-function description (K02).

Rows where neither option fits well

- **Row 1 (Surface tension):** K03 covers the *quantum* physics of matter, but surface tension is taught classically. "None" (option A) was chosen, although the item's place in S5's Physical Foundations list leaves K03 arguable.
- **Row 4 (Electrons):** The bare noun fits K04 literally (a subatomic particle) and K03 only through context (atoms). Electrons treated as quantum objects could also fall under K01.

Rules applied for consistency

- **S4:** Where an item fits both options, S4's own grouping decides: the key-concept category in Table 2, or the key-concept tables (8a–8i) in which an illustration is offered. This covers rows 2, 3, 8, 14, 16, 20, 21 and 22.
- **S3:** Table 6 is sorted by the number of experts, so adjacent items share a vote count, not a theme. They were not used as evidence.
- **S5:** Tables III–V list topics "in sequence as given to the experts", so adjacent items were used as evidence (rows 4, 5, 9 and 17).
- **S6:** The source's own thematic grouping (Table III) and its nature-of-science framing (Table I) were used for rows 13 and 19.

Neutral re-run of S6-018: pre-registration and result

- Decided by the authors on 2026-10-26 after a model-run review of the revised manuscript noted that the original instructions singled this item out.
- Item: S6-018, "Methodology (e.g., experiments and hypothesis)", Stadermann et al. 2019, Table I aspect N1.
- Original run (adjudication_run/, 2026-10-26): decision K12 (nature of science), confidence high. The original instructions contained a paragraph naming this item as possibly ambiguous and stating the two readings.

- This run: same model (Claude Opus 5.5, claude-opus-5-5), same effort setting (max), fresh agent; same cluster definitions and source texts; the instructions are the original ones with that paragraph removed and the counts changed from 26 items to one. Option order re-randomised (seed 20261027): option_A = K12, option_B = K13 (the original run had A = K13, B = K12).
- The agent is not shown the original decision and is told not to open anything outside this folder.
- Pre-registered reporting: both decisions are reported in §3.3 and S4. If this run decides K13 (practices), R-crosscut is re-applied mechanically and its result reported; whether practices then become a domain is put to the authors before any manuscript change. If it decides K12, the text states that the neutral re-run agreed.

Result (added after the run)

- 2026-10-26T15:28:35Z: decision K12 (option A in this run's order), confidence high — the same cluster as the original run. Under the pre-registered reporting, the text states that the neutral re-run agreed; R-crosscut is unchanged (practices remain a layer).

Neutral re-run of S6-018: summary

- Item: S6-018, "Methodology (e.g., experiments and hypothesis)".
- Decision: A = K12 Nature, history and philosophy of quantum physics. Confidence: high.
- Reason (from DECISIONS_MODEL.csv): "NOS aspect on scientific method (experiment-theory relation, thought experiments in QP's development): K12 'what it shows about the nature of science', not K13 'activities that learners carry out'."
- Run: 2026-10-26T15:28:35Z (progress.log). PACKET_FOR_MODEL.csv unchanged (md5 edceef90d5f158c2ccedc63eefbc3c29).